Enhancing Malware Classification with Vision Transformers: A Comparative Study with Traditional CNN Models

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ABSTRACT

Malware classification is an important task in cybersecurity, where machine learning techniques have been widely used to automate the process of identifying and categorizing malicious software. In this paper, we propose a new approach to malware classification using Vision Transformers (ViT), a state-of-the-art deep learning architecture that has shown good results in computer vision field. Our proposed ViT-based model outperforms traditional Convolutional Neural Networks (CNNs) in terms of accuracy and robustness, as it can capture long-range dependencies in the input data without relying on hand-crafted features. We evaluate our proposed model on Malimg database, and demonstrate that it achieves state-of-the-art performance, outperforming the existing traditional approaches. Furthermore, we investigate the impact of different input representations, model configurations, and training strategies on the ViT-based model's performance. Our results show that the proposed ViT-based model offers several advantages over traditional CNN models, such as better performance on large-scale and complex datasets, higher interpretability, and scalability. However, the ViT-based model requires significantly more computational resources and longer training time. Our proposed approach offers a promising direction for malware classification using ViT-based models, which can be further improved by exploring different architectures, optimization techniques, and transfer learning strategies.

CCS CONCEPTS

• Security and privacy; **• Intrusion/anomaly detection and malware mitigation**; **• Malware and its mitigation.**;

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KEYWORDS

Cybersecurity, Malware classification, Vision Transformers, Convolutional Neural Network

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1 INTRODUCTION

Malware, short for malicious software, refers to any program or code designed to cause harm to computer systems, networks, or mobile devices. Malware can be used for a variety of purposes, including theft of personal information, financial fraud, and espionage. Malware attacks are becoming increasingly common and sophisticated, making it essential for individuals and organizations to have effective malware detection and classification systems in place. According to recent statistics, the number of malware attacks has been steadily increasing. In 2020, there were over 160 million new malware samples detected, which represents a 358% increase from the previous year. In addition, the average cost of a data breach caused by malware was $4.27 million in 2020. These alarming figures highlight the critical need for effective malware detection and classification strategies. Today, in 2023, New malware detected in the last 14 days is 2,603,943 new malware by AV-TEST institution statistics [1].

Malware detection and classification involve identifying and analyzing suspicious programs or codes to determine whether they are malicious or not. This process is crucial in preventing malware attacks and minimizing their impact when they occur. Effective malware detection and classification systems can help individuals and organizations stay protected against new and emerging threats and can significantly reduce the financial and reputational damage caused by a successful malware attack. The importance of malware detection and classification cannot be overstated. With the increasing number and complexity of malware attacks, individuals and organizations need to invest in effective detection and classification systems to stay protected against these threats. To combat these challenges, many organizations are adopting advanced malware detection and classification technologies, such as machine learning and behavioral analysis. These technologies can help identify and classify new and unknown malware, enabling organizations to stay ahead of emerging threats.

On the other hand, in computer vision, Convolutional neural networks (CNN) have been so successful and powerful, such as: AlexNet, ResNet and others. In 2021, Researchers applied transformers to image classification [2]. And, it comes with no surprise that transformers outperform CNN showing that the reliance on CNNs is not necessary anymore. The direct application of a pure transformer to sequences of images patches performs very well on image classification tasks. Also, transformers require fewer computational resources to train models [3].

Transformers operates many operations on the sequences [4]. For instance, if we have a set of words, the transformer would take all of these in and do more computations with them which is attention. Attention is a quadratic operation which needs to calculate the pairwise inner products between each pair of the sequences data. This becomes a very large task very quickly because there are many interconnections. For example, in NLP, if a paragraph is 500 tokens long then we need 500 squared connections. So, this is one of the limitations of transformers. Transformers work very well for NLP [5], however, they are limited by the memory and compute requirements of the quadratic attention. Images are much harder for transformers, because an image is a raster of pixels. A normal image has many pixels in the range of 1000-2000 pixels side length to seems clear for humans. The rasterization of images is a problem in itself even for CNNs then feeding this to transformers every single pixel has to attend to every single other pixel. If we have a 250x250 pixels image, the image itself is 2502 then the attention will be (2502)2 which is impossible to handle in current hardware. To deal with this, in this moment, we're resorting to only local attention. Transformers are able to attend within a single layer to everywhere.

Vision transformer requires partitioning the input image into patches of the same shape for example we can partition the image into 9 patches the same shape where they don't have overlap. However, the input image can be split into overlapping patches, it's up to the user's perspectives. When splitting the image, we use the sliding window that moves several pixels each time. Stride is the number of pixels the sliding window moves each time. A smaller stride gives a bigger number of patches. To split an image, we need to specify two arguments: the patch size and the stride. Using transformers, the patch size is 16x16 patches. Our image is split into non-overlapping patches, so the stride is also 16x16. Every patch is a small image with RGB channels. A patch is an order of 3 tensors. Then, these patches are vectorized. Vectorization means reshaping a tensor into a vector. The 16 tensors are reshaped into 16 vectors [6].

In this paper, we introduce a novel approach to classify malware using Vision Transformers (ViT), a deep learning architecture that has shown promising results in computer vision tasks. By leveraging ViT-based models, we are able to outperform traditional Convolutional Neural Networks (CNNs) in terms of accuracy and robustness. Our experimental results demonstrate that the ViT-based model achieves state-of-the-art performance in term of malware classification using visualization technique. We also investigate the impact of different input representations, model configurations, and training strategies on the ViT-based model's performance. Compared to traditional CNN models, ViT-based models offer several advantages, such as better performance on large-scale and complex datasets, higher interpretability, and scalability. However, the ViT-based model requires significantly more computational resources and longer training time. This paper presents a promising direction for malware classification using ViT-based models, which can be further improved by exploring different architectures, optimization techniques, and transfer learning strategies.

2 LITERATURE REVIEW

In 2017, the Transformer Neural Network architecture [7] was introduced. This network employs an encoder decoder architecture, much like RNNs. The difference between RNNs and transformers is that the input sequence can be passed in parallel. Retaking the example of translator, using RNN encoder we give an input Arabic sentence one word after the other. The current words hidden state has dependencies in the previous words hidden state. The word embeddings are generated one time step at a time. On the other hand, using transformer encoder there is no concept of time step for the input. We give all the words of the sentence simultaneously and determine the word embeddings simultaneously. Today, TensorFlow provides in latest versions (+2.5) the implementation of transformers and its different components. They also give a step-by-step tutorials series with detailed explanation regarding highly NLP field. Transformer neural networks have largely replaced LSTM networks for sequence-to-vector, sequence-to-sequence and vector-to-sequence problems. In 2019, Google created BERT [8] which uses transformers to pre-train models for common NLP tasks.

2.1 Multi-head attention layer components

The single head attention contains Q, K and V which are abstract vectors that extract different components of an input word. We have Q, K and V vectors for every single word. We use these to compute the attention vectors for every word using Z formula (eq.1).

$$Z = Attention\ (Q,\ K,\ V) = softmax(\frac{Q.K^T}{\sqrt{Dimension\ of\ vector\ Q,\ K\ or\ V}}).V \tag{1}$$

For multi-head attention, we have multiple weight matrices WQ, WK and WV then we have multiple attention vectors for every word. However, the next feed-forward network is only expecting one attention vector per word, so we use another weighted matrix WZ to make sure that the output is still an attention vector per word.

$$W^Z = MultiHead(Q,\ K,\ V) = Concat(head_i,\ \dots,\ head_n).\ W^O \tag{2}$$

where :

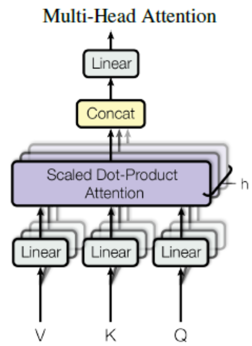

Figure 1: multi-head attention architecture [7]

$$head_i = Attention(\ WQ_i^Q,\ WK_i^K,\ WV_i^V)$$

2.2 Malware visualization technique : Malimg database

In many cases, a malware is identified as a Portable Executable (PE) file. However, the task of classifying malware has been transformed into an image classification problem by representing binary executables as grayscale or colored images, presented as arrays in the range of 0 to 255. This approach was first introduced by Nataraj in 2011 [9], who created the Malimg database containing 9369 malware images in 25 classes [10]. Since then, this technique has been widely adopted in various works that aim to effectively classify malware samples into their respective families. The technique of representing malware as images for classification has become increasingly important in the field of cybersecurity. This approach allows for the use of powerful image processing and machine learning algorithms to detect and classify different types of malwares, including new and unknown variants. By visualizing malware as images, it becomes easier to identify patterns and features that can be used to distinguish one type of malware from another. This technique also enables the creation of large datasets of labeled malware images, which can be used to train and test machine learning models for malware classification. As malware continues to evolve and become more sophisticated, this technique provides a valuable tool for detecting and preventing cyberattacks.

3 PROPOSED SOLUTION

In recent years, there has been an increase in the number of malware attacks and the complexity of malware variants. This has led to a greater need for more efficient and effective methods for malware classification. Researchers have explored various techniques for classifying malware samples, including traditional machine learning algorithms and deep learning models. The success of ViT (Vision Transformer) on vision tasks, such as image classification, has motivated researchers to apply this approach to malware classification. The visual similarity between malware samples of the same family suggests that ViT may also be effective in classifying malware based on their visual features. Therefore, in this study, we propose MalwareViT, a method for applying Vision Transformers to malware classification. MalwareViT has shown promising results in experiments and has the potential to improve the efficiency and accuracy of malware classification in the future.

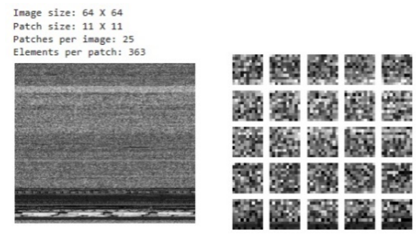

Figure 2: Plot patches sample from Malimg database

The architecture of a Vision Transformer consists of a stack of Transformer layers, which take in image patches (fig. 2) as input and produce class probabilities as output. The ViT model architecture consists of six blocks as given in figure 3. Each Block, from 1 to 6, is composed of Layer normalization and Multi Head attention layer, in addition to Layer normalization, dense and dropout layers. This block of attention is been repeated 6 times. Then, we use another layer normalization and flattens them into 1D vectors. The final block series is a classification layer that takes the output of the previous block and maps it to the output classes. At this stage we use the DD block, which contains a Dropout and a Dense layer. The DD Block is repeated 3 times before giving the malware family prediction result.

The performance of a Vision Transformer model depends heavily on the choice of hyperparameters (table 1). One important hyperparameter is the number of Transformer layers, which affects the depth and capacity of the model. Another important hyperparameter is the patch size, which determines the granularity of the input image. Additionally, the size of the model's hidden state, the number of attention heads, and the dropout rate are also important hyperparameters to consider. Finding the optimal hyperparameter values for a Vision Transformer model is a crucial step in achieving state-of-the-art performance on image classification tasks.

4 RESULTS AND DISCUSSION

It is difficult to say definitively which type of model, Convolutional Neural Networks (CNNs) or Vision Transformers (ViT), is better for image recognition tasks, as the choice of model will depend on the specific characteristics of the task and the available resources. Both CNNs and ViT have their own strengths and weaknesses, and each may be better suited for certain types of tasks or scenarios. One of the main advantages of CNNs is that they have been widely studied and are well understood, and they have been shown to be very effective for a wide range of image recognition tasks. CNNs are also relatively efficient to train and deploy, and they can be trained on relatively small datasets. On the other hand, ViT are a relatively new type of model and they have been shown to achieve state-of-the-art performance on a number of image recognition benchmarks, including the ImageNet dataset. ViT are particularly well-suited for processing images of any size, and they can handle input sequences of arbitrary length in a parallel and efficient manner.

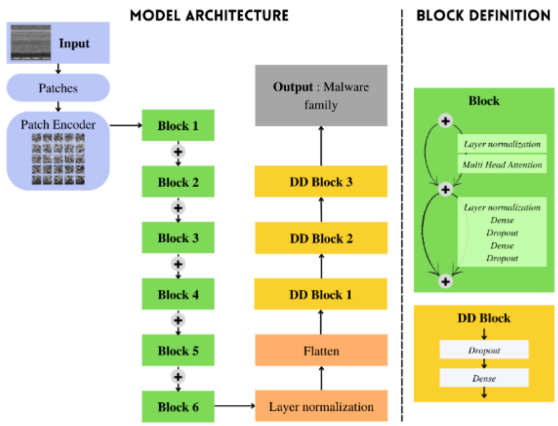

Figure 3: Malware Vision Transformer model architecture

Table 1: Vision transformer hyperparameters

Classes	25
Input shape	(64,64,3)
Learning rate	0.001
Weight decay	0.0001
Batch size	256
N° of epochs	100
Patience: After patience epoch stop if not improve	30
Image size	64
Patch size: Size of the patches to be extract from the input images	11
N° of patches = (image_size // patch_size) ** 2	25
Projection dim	36
N° of heads	4
Transformer units (Size of the transformer layers) = [projection_dim * 2, projection_dim]	(72, 36)
Transformer layers	8
MLP head units	[2048, 1024]

Table 2: Results comparison

Model	Metrics	
	Accuracy	Loss
ViT	0.9838	0.0517
CNN	0.9790	0.1229
KNN+GIST [11]	0.9737	-0.1627 (Negative log loss)

In our experimentation, we found the results presented in figure 9 and table 2. The loss function decreases by epochs time to a very low values during training and validation phases. Also, accuracy increases to high values during training epochs. Using the proposed ViT architecture presented previously, we reach 98.38% of accuracy, and we have a 5.17% of loss. These results are very close to the literature. However, the particularity of ViT model is being much concise and accurate. In order to make the comparative study, we have built a CNN classifier to compare results. Our best CNN classifier for malware classification task reach 97.9% of accuracy and 12.29% of loss. It is true that in literature we could find greater accuracy, but the model architecture complexity is also a criterion to take into consideration. And to have a complete comparative study we have also compare with classical machine learning algorithms in the same context, where the k-Nearest Neighbor malware classifier gives only 97.37% of accuracy and 16.27% of log loss.

Ultimately, the choice between CNN and ViT will depend on the specific requirements of the task at hand, as well as the available

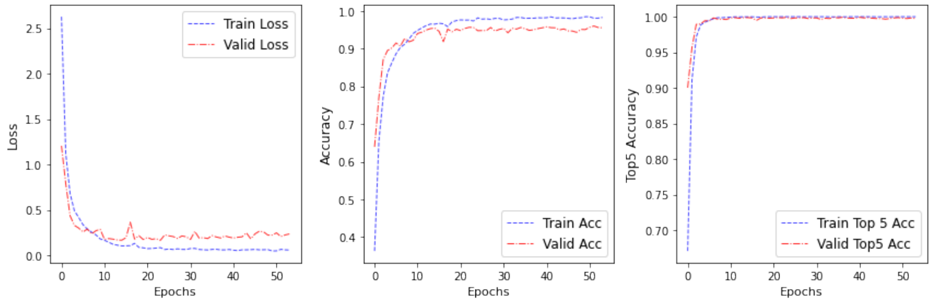

Figure 4: ViT Model training results

resources and constraints. It may be necessary to try out both types of models and compare their performance to determine which one is the best fit for a particular task. For malware classification task using images we recommend the use of ViT for better and precise performance.

5 CONCLUSIONS

In conclusion, this paper presents an effective approach for malware classification using Vision Transformers model (ViT), a novel deep learning architecture that has shown promising results in various computer vision tasks. The proposed ViT-based model outperforms traditional Convolutional Neural Networks (CNNs) in terms of accuracy and robustness, as it can capture long-range dependencies in the input data without relying on hand-crafted features. The experimental results demonstrate that the ViT-based model achieves state-of-the-art performance on Malimg database. Furthermore, the paper also investigates the impact of different input representations, model configurations, and training strategies on the performance of the ViT-based model.

Compared to traditional CNN models, the ViT-based model has several advantages, such as improved performance on large-scale and complex datasets, higher interpretability, and scalability. However, it requires significantly more computational resources and longer training time due to its larger model size and complex attention mechanism. Overall, this paper presents a promising direction for malware classification using ViT-based models, which can be further improved by exploring different architectures, optimization techniques, and transfer learning strategies. Nevertheless, the traditional CNN models still remain a viable option for malware classification, especially for smaller and less complex datasets, where they can achieve competitive performance with less computational cost.

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